Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

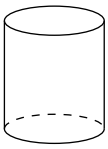
Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Cylinders and Spheres: Their Properties

Two three-dimensional shapes you will meet often are the cylinder and the sphere. Look at fig-01: a cylinder drawn upright. It has a flat circular top and a flat circular base, with a curved wall joining them. A cylinder has these properties:

1. It has two circular faces (the top and the base) and one curved face.
2. It has no vertex — there is no sharp corner anywhere.
3. It has two curved edges, where each circular face meets the curved wall.
4. It has three faces altogether — the two flat circular faces and the one curved face.

Visual

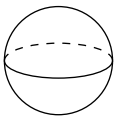


A cylinder drawn upright: an ellipse for the top face, two vertical sides falling from its ends, and a curve closing the base.

Now look at fig-02: a sphere, drawn as a circle with a curved line across its middle to show that its surface is curved. Think of a ball — that is a sphere. A sphere has these properties:

1. It is symmetrical from every direction — whichever way you turn it, it looks the same.
2. It has no edge and no vertex — there is nothing sharp on a sphere at all.
3. It has one curved surface only.

Visual



A sphere drawn as a circle with an ellipse across its middle to show the curved surface.

Practice

1. How many circular faces does a solid cylinder have?
2. A cylinder has how many faces altogether?
3. How many vertices does a cylinder have? How many vertices does a sphere have?

Section 2: Finding the Volume of Cubes, Cuboids and Prisms

Volume is the amount of space a three-dimensional shape takes up, and we measure it in cubic units such as cm^3, m^3 or mm^3. To find the volume of any solid that has a flat base and straight vertical sides, we can multiply the area of the base by the height of the shape.

For a cuboid, the base is a rectangle, so the area of the base is length × breadth. Multiplying that by the height gives the volume:

Volume of a cuboid = length × breadth × height = lbh

A cube is a special cuboid where all three edges are the same length, so length = breadth = height = l. Its volume is therefore:

Volume of a cube = length × length × length = l^3

Example 1. Calculate the volume of a cuboid of length 3 cm, breadth 2 cm and height 8 cm.

Solution:

Volume of a cuboid = length × breadth × height

length = 3 cm, breadth = 2 cm, height = 8 cm

Volume = 3 cm × 2 cm × 8 cm = 48 cm^3

So the volume of the cuboid is 48 cm^3.

Example 2. The volume of a cuboid is 24 cm^3. If the base is 6 cm by 2 cm, find the height of the cuboid.

Solution:

Volume of a cuboid = length × breadth × height

length = 6 cm, breadth = 2 cm, height =?

Volume = 24 cm^3

So 24 cm^3 = 6 cm × 2 cm × h

24 cm^3 = 12 cm^2 × h

To find h, divide both sides by 12 cm^2:

24 cm^3 ÷ 12 cm^2 = h

h = 2 cm

So the height of the cuboid is 2 cm.

Example 3. Calculate the volume of a cube of side 3 cm.

Solution:

Volume of a cube = l^3

V = (3 cm)^3 = 3 cm × 3 cm × 3 cm = 27 cm^3

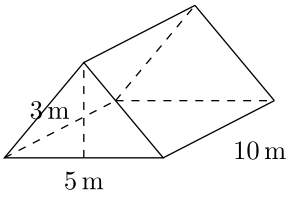
So the volume of the cube is 27 cm^3.

A prism is a solid with a uniform cross-section, meaning the same shape runs all the way along its length. For example, the prism in fig-03 is a triangular prism: its end is a triangle, and that triangle runs back 10 m along the prism. Its base does not have to be a rectangle, so we use the general rule:

Volume of a prism = area of base × height

Here "base" means the cross-section that is repeated, and the "height" is how far that shape runs along the prism.

Visual



A triangular prism lying on its rectangular face. The triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

Example 4. In fig-03, the triangular prism has a triangular end with a base of 5 m and a height of 3 m, and the prism runs back 10 m. Calculate its volume.

Solution:

First find the area of the triangular base.

Area of a triangle = ½ × base × height = ½ × 5 × 3 = 7.5 m^2

Now multiply by the distance the prism runs back:

Volume of prism = area of base × height = 7.5 m^2 × 10 m = 75 m^3

So the volume of the prism is 75 m^3.

Practice

1. Calculate the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.
2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
3. What is the volume of a cube with sides of 4 cm?
4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 m^3 of water. How deep is the water in the tank?

Check your work

Attempt every question before you read this page.

Cylinders and Spheres: Their Properties

1. A solid cylinder has 2 circular faces.

   A solid cylinder has a circular top and a circular base, so it has 2 circular faces.
2. 3 faces — the two circular faces and the one curved face.

   Count the two flat circular faces plus the one curved face: $2 + 1 = 3$ faces altogether.
3. A cylinder has no vertex, and a sphere also has no vertex.

   Neither shape has a sharp corner, so neither has a vertex.

Finding the Volume of Cubes, Cuboids and Prisms

1. 96 cm^3

   Volume = length × breadth × height = 8 cm × 4 cm × 3 cm = 96 cm^3.
2. 30 m^3

   Volume = length × breadth × height = 5 m × 2 m × 3 m = 30 m^3.
3. 64 cm^3

   Volume of a cube = $l^3 = (4\text{ cm})^3 = 4\text{ cm} \times 4\text{ cm} \times 4\text{ cm} = 64\text{ cm}^3$.
4. 3 m

   Volume of water = length × breadth × height, so $36\text{ m}^3 = 6\text{ m} \times 2\text{ m} \times h = 12\text{ m}^2 \times h$. Dividing both sides by 12 m^2 gives $h = 36\text{ m}^3 \div 12\text{ m}^2 = 3$ m. So the water is 3 m deep.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-e8d88da8	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____